

OSSAMA TAHA

BI & DATA ANALYTICS ENGINEER

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SUMMARY

BI & Data Analytics professional with a B.Com in Accounting and 2+ years of experience building production-grade data pipelines, automated reporting systems, and executive dashboards. Proven track record of reducing reporting cycles by 10x through end-to-end automation, enabling data-driven strategic decisions for sales and operations teams. Combines deep business domain knowledge with strong technical skills in Python, SQL, PostgreSQL, Apache Airflow, dbt, and cloud infrastructure.

EXPERIENCE

- Nile for Cables and Electronics, BI & Sales Data Analyst

- Reduced reporting cycle from days to hours by designing and deploying a semi-realtime automated reporting system, directly improving sales team responsiveness and strategic decision-making
 - Led quarterly sales performance evaluations using consolidated KPIs, providing actionable insights that drove revenue-focused business decisions
 - Built executive dashboards supporting sales forecasting and real-time KPI monitoring, enabling stakeholders to track performance without manual data pulls
 - Designed end-to-end ETL pipelines using PostgreSQL and Dolt (version-controlled SQL database), ensuring data integrity and audit trail for critical financial metrics
 - Developed a local AI chatbot using open-source LLMs for natural language querying of business data, empowering non-technical stakeholders to self-serve analytics

Giza, Egypt

Nov 2024 – present

1 year 7 months
- Samsung Innovation Campus, Data Engineer

- Developed scalable data processing workflows using Apache Spark and Kafka, supporting business intelligence and analytical use cases
 - Performed exploratory data analysis and feature engineering to enable data-driven business insights
 - Deployed production data workflows on AWS cloud infrastructure (EC2, S3), ensuring reliable and cost-efficient operations

Giza, Egypt

July 2024 – Oct 2024

4 months

PROJECTS

- dbt Data Warehouse – Star Schema & Analytics Layer

- Problem:** Raw operational data lacked structure for analytics, forcing manual data wrangling for every report
 - Process & Solution:** Designed star schema with 15+ dbt models across staging/marts layers, SCD Type 2 tracking, incremental processing, and 99.9% data quality via automated testing
 - Business Impact:** Reduced query complexity by 70%, eliminated manual data prep, enabled self-service analytics
- Atlas Analytics Platform

- Problem:** Siloed data across departments required manual updates with no unified visibility
 - Process & Solution:** Built full-stack platform with Python backend, automated ETL pipelines, role-based dashboards, deployed on AWS with CI/CD
 - Business Impact:** Enabled data-driven decisions across 3 departments, reduced manual reporting by 80%
- Airflow ETL Pipeline – Multi-Source Data Orchestration

- Problem:** Manual API data extraction was error-prone with no scheduling or failure recovery
 - Process & Solution:** Designed 4 production DAGs with task dependencies, dynamic mapping, retry logic, extracting from 3 API sources into PostgreSQL with real-time monitoring
 - Business Impact:** Fully automated daily/weekly ingestion with zero manual intervention
- Agentic AI Workflow – Multi-Agent DataOps Automation

- Problem:** Data operations required constant manual monitoring and debugging across multiple systems
 - Process & Solution:** Built multi-agent system with orchestrator pattern, 5 specialized agents, tool integration, and persistent memory using LangChain/LangGraph
 - Business Impact:** Cut pipeline failure resolution time by 70%, automated 90% of routine monitoring
- AWS Cloud Data Pipeline – Serverless ETL with IaC

- Problem:** On-premise infrastructure lacked scalability and cost efficiency for growing data
 - Process & Solution:** S3 data lake + Glue ETL + Athena queries + Step Functions orchestration, fully deployable via Terraform IaC
 - Business Impact:** 90% cost reduction vs traditional infra, sub-second query performance, zero server management
- Kafka Streaming Pipeline – Real-Time Data Processing

- Problem:** Batch processing couldn't meet real-time analytics needs for stock monitoring and telemetry
 - Process & Solution:** Event-driven architecture with Kafka producers, consumer groups, windowed aggregation, dead letter queues, live dashboard, and PostgreSQL sink
 - Business Impact:** Sub-second data latency, automated threshold alerting, real-time operational visibility

SKILLS

- BI & Data Visualization:

Power BI, DAX, Tableau, Excel, Streamlit, Plotly, Dash

Data Pipeline & Architecture:

ETL/ELT Pipelines, Apache Airflow, dbt, Apache Spark, Kafka, Data Modeling (Star Schema), PostgreSQL, MySQL

Cloud & Infrastructure:

AWS (S3, Glue, Athena, Lambda, Step Functions, EC2), Terraform, Docker, Docker Compose, Linux Server Administration, Nginx, GitHub Actions (CI/CD)

Programming & Scripting:

Python, SQL, Bash, JavaScript, TypeScript, HTML/CSS

AI & Machine Learning:

LangChain, LangGraph, Hugging Face Transformers, Scikit-learn, TensorFlow, NLP, Local LLM Deployment

Data Quality & Monitoring:

Great Expectations, Data Validation, Schema Testing, Anomaly Detection, Pipeline Monitoring

Automation & Tools:

Git, n8n, OpenClaw, KiloCode, Hermes, Dolt, Jupyter, VS Code

EDUCATION

Al-Azhar University, Accounting & Business Management

Sept 2019 – June 2023

Bachelor of Commerce

- Self-taught in programming, data engineering, and cloud technologies through professional certifications and hands-on projects

CERTIFICATIONS

Google Data Analytics Professional Certificate

- Data cleaning, analysis, visualization with spreadsheets, SQL, Tableau, and R

IBM Data Engineer Professional Certificate

- ETL pipelines, data warehousing, big data (Spark, Hadoop, Kafka), cloud deployment

IBM Data Science Professional Certificate

- Machine learning, data analysis, Python, SQL, applied data science capstone projects

Samsung Innovation Campus Big Data Engineering Certificate

- "Top Achiever (#1) in the Big Data Engineering track"
- Intensive training in big data infrastructure, distributed computing, and data engineering

AWS Certified Data Engineer – Associate

- S3, Glue, Athena, Lambda, Step Functions, and Terraform (In Progress)

iMBA Business Analysis Specialization (In Progress)

- Coursera iMBA from UIUC – business process modeling, requirements engineering, data-driven decisions